Coordinate frame definitions and transformations for polarisation analysis using `PolPy`

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1 Introduction

The purpose of this document is to define the standard coordinate frames (and transformations between them) needed to carry out the polarization analysis within the `PolPy` framework. The document is arranged in the following format: Section 2 describes all the symbols and the coordinate frame definitions used and Section 3 describes the transformations between them.

2 Definition of frames and the polarization angle

There are four frames, the celestial reference frame (J2000), the instrument frame (IRF), the local tangent plane frame (LTP) and the projected celestial reference frame at the source location in which the polarization angle is defined according to the IAU convention (IAU). The following subsections define the frames along with Figures 1 and 2.

2.1 The celestial frame (J2000)

The celestial frame is defined in the standard IAU convention where the Z-axis points to the Celestial North pole and the X-axis points to the Vernal Equinox at a given Epoch, in this case J2000, which is fixed at noon on January 1, 2000. A direction in the frame is defined by Right Ascension (RA, α) and Declination (Dec, δ).

2.2 The Instrument Reference Frame (IRF)

The definition of the instrument reference frame varies for each instrument but typically the Z-axis (or also called the pointing axis) of the frame points towards the Zenith of the detector plane and the X and Y axes are in the plane of the detector plane. Typically, the Z-axis points towards a certain direction (α, δ) in the J2000 frame. Note that some instruments may use the X or Y axis as the pointing direction.

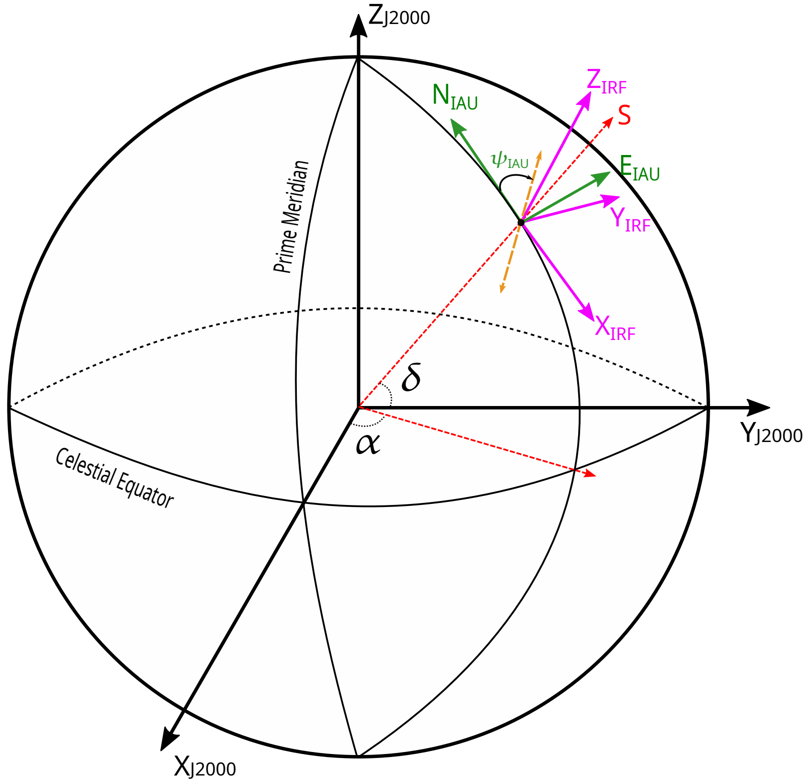

Figure 1: A diagram depicting the relation between J2000 and IRF and the definition of Polarization Angle in the celestial frame. The three basis vectors of J2000 are shown in black, the source direction (α, δ, and its projection into the XY plane) is shown by the dotted red lines. The IRF basis vectors are shown in magenta, the Z-axis here marks the instrument pointing direction. The green vectors show the local (with respect to the celestial North pole) North and East in the tangent plane perpendicular to the source direction. The orange dotted line marks the projection of the electric field vector and the polarization angle is defined as ψ_{J2000}.

2.3 The Local Tangent Plane frame (LTP)

The local tangent plane frame is defined in the plane perpendicular to the source direction[1] relative to the IRF. The Z-axis points to the direction opposite to the source direction, the X-axis points to the "local" North with respect to the IRF and the Y-axis points to the "local" East with respect to the IRF.

2.4 The IAU Polarization Angle (PA) frame

The IAU Convention (year 1973, commission 40, page 211) states that polarization angle should be measured from local North increasing positively towards the local East in the celestial frame. We define this frame as the IAU frame for following discussion. Note that for a given direction LTP and IAU share the same XY plane but the X axes of both frames have an offset, i.e. they are rotated around the source direction. The aim of the transformation is to find the offset between

[1]The LTP is the same as the standard North-East-Down (NED) frame defined in aviation, see https://en.wikipedia.org/wiki/Local_tangent_plane_coordinates for more details. To avoid confusion with celestial North and East, LTP convention is used here

the two X axes and apply it to the polarization responses which are computed in the LTP to transform them into the IAU frame.

3 Frame transformations

To compute the offset between IAU and LTP we need to carry out two transformations, one to transform the polarization angle reference axis in IAU, N_{IAU}, to the IRF and then transform it further to the LTP. The first matrix is defined as $R_{J2000 \rightarrow IRF}$ and the second one is defined as $R_{IRF \rightarrow LTP}$. The final transformation matrix is the product of the two matrices, $R_{J2000 \rightarrow LTP} = R_{IRF \rightarrow LTP} \times R_{J2000 \rightarrow IRF}$. The discussion below defines all the matrices used in the `PolPy` code in detail.

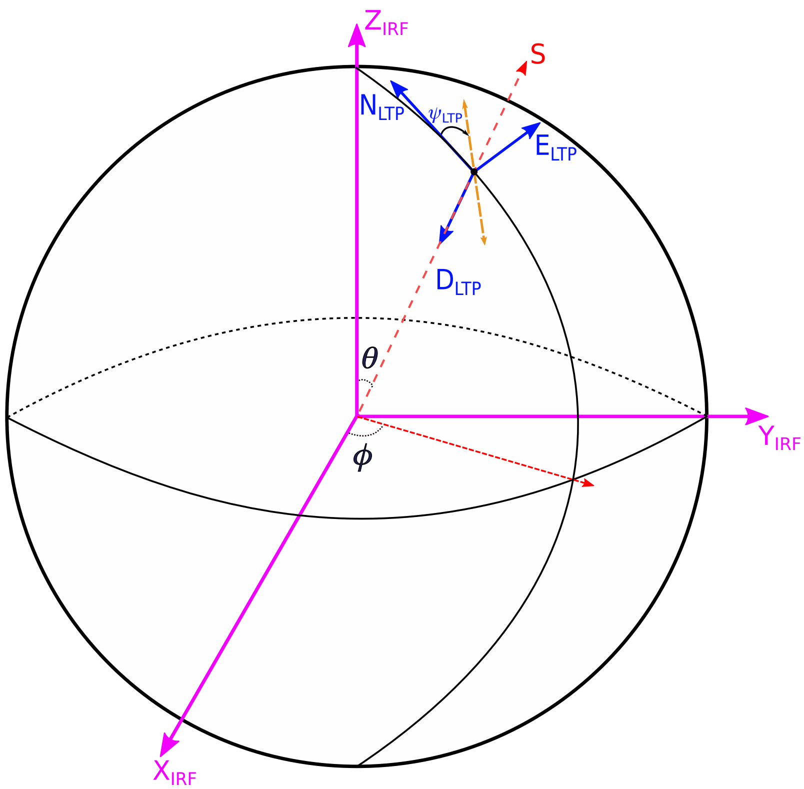

Figure 2: A diagram showing the relation between IRF and LTP and definition of polarization angle in the LTP. The IRF basis vectors are shown in magenta and the source direction in IRF (θ, ϕ, and its projection into the XY plane) is shown by the dotted red lines. The LTP basis vectors are shown in blue. As before, IRF Z-axis direction corresponds the pointing direction of the instrument. The Z-axis of LTP points in the direction opposite to the source (i.e. the direction of incoming photons). The X and Y axis of LTP are the local (with respect to instrument North, i.e. the pointing direction) North and East. The orange dotted lines marks the projection of electric field vector and the Polarization Angle in LTP is defined by ψ_{LTP}

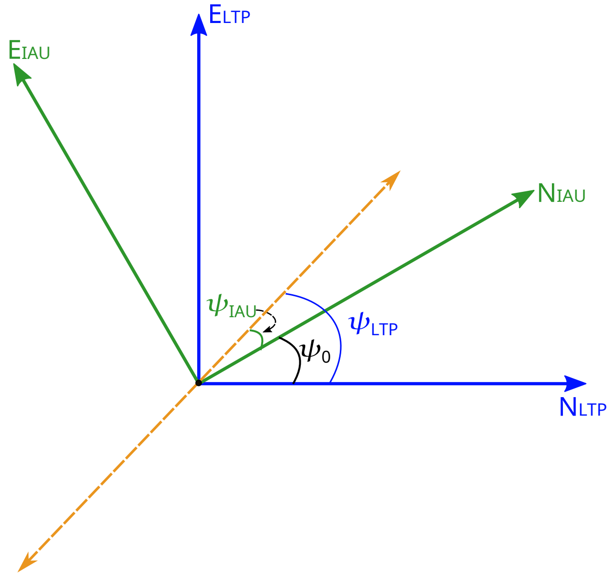

Figure 3: Projected view of the tangent plane perpendicular to the source direction as seen from the source. The orange dotted line is the projected electric field vector. As defined by the IAU convention the ψ_{J2000} is measured from the local North increasing positively towards the East. The ψ_{LTP} is defined in a similar way but with respect to the LTP North and East.

3.1 J2000 to IRF

Given the RA (α) and Dec (δ) of the three instrument axes X, Y and Z, the transformation matrix to go from IRF to J2000 can be written as:

$$R_{J2000\rightarrow IRF} = \begin{bmatrix}\vec{X} & \vec{Y} & \vec{Z}\end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \cos\alpha_{\mathrm{X}}\cos\delta_{\mathrm{X}} & \sin\alpha_{\mathrm{X}}\cos\delta_{\mathrm{X}} & \sin\delta_{\mathrm{X}} \\ \cos\alpha_{\mathrm{Y}}\cos\delta_{\mathrm{Y}} & \sin\alpha_{\mathrm{Y}}\cos\delta_{\mathrm{Y}} & \sin\delta_{\mathrm{Y}} \\ \cos\alpha_{\mathrm{Z}}\cos\delta_{\mathrm{Z}} & \sin\alpha_{\mathrm{Z}}\cos\delta_{\mathrm{Z}} & \sin\delta_{\mathrm{Z}} \end{bmatrix} \tag{1}$$

Note that in a real case, to find $R_{IRF\rightarrow LTP}$, knowing α and δ of $\vec{X}$ and $\vec{Z}$ is enough and $\vec{Y}$ can be estimated as:

$$\vec{Y} = \vec{Z} \times \vec{X} \tag{2}$$

3.2 LTP to IRF

Given the θ and ϕ of the source, the direction vectors of the X, Y and Z axes of LTP in IRF are given as:

$$X_{LTP} = \begin{bmatrix} \sin(90-\theta)\cos(\phi+180) \\ \sin(90-\theta)\sin(\phi+180) \\ \cos(90-\theta) \end{bmatrix} \quad \text{for} \quad \theta < 90^\circ; \quad X_{LTP} = \begin{bmatrix} \sin(\theta-90)\cos\phi \\ \sin(\theta-90)\sin\phi \\ \cos(\theta-90) \end{bmatrix} \quad \text{for} \quad \theta > 90^\circ \tag{3}$$

$$Y_{LTP} = \begin{bmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{bmatrix} \quad Z_{LTP} = \begin{bmatrix} -\sin\theta\cos\phi \\ -\sin\theta\sin\phi \\ -\cos\theta \end{bmatrix} \tag{4}$$

Hence, the transformation matrix to go from IRF to LTP can be written as:

$$R_{IRF \to LTP} = \begin{bmatrix} X_{LTP} & Y_{LTP} & Z_{LTP} \end{bmatrix}^{\mathrm{T}} \tag{5}$$

3.3 Polarization angle offset computation

As mentioned before, the polarization angle offset (ψ_0) is computed by transforming the IAU reference axis to the LTP and then finding the angle between the two X axes. At any given source direction α_S, δ_S, the reference axis for PA in the IAU frame (or the direction of the local North) is given as:

$$N_{IAU} = \begin{bmatrix} \cos(90-\delta)\cos(\alpha+180) \\ \cos(90-\delta)\sin(\alpha+180) \\ \sin(90-\delta) \end{bmatrix} \quad \text{for} \quad \delta > 90 \tag{6}$$

$$= \begin{bmatrix} \cos(90+\delta)\cos\alpha \\ \cos(90+\delta)\sin\alpha \\ \sin(90+\delta) \end{bmatrix} \quad \text{for} \quad \delta < 90 \tag{7}$$

The projection of N_{IAU} in the XY plane of LTP is given as:

$$\mathrm{N}_{\mathrm{IAU}}^{(\mathrm{LTP})} = R_{J2000 \to LTP} \times N_{IAU} \tag{8}$$

$$= R_{IRF \to LTP} \times R_{J2000 \to IRF} \times N_{IAU} \tag{9}$$

$$\text{and} \quad \psi_0 = \mathrm{arctan2}\left(\frac{N_{\mathrm{IAU}}^{(\mathrm{LTP})}(y)}{N_{\mathrm{IAU}}^{(\mathrm{LTP})}(x)}\right) \tag{10}$$

And the PA in the IAU frame (ψ_{IAU}) is related to the PA in the LTP frame (ψ_{LTP}) as:

$$\psi_{\mathrm{IAU}} = \psi_{\mathrm{LTP}} - \psi_0 \tag{11}$$

$$= (180 + \psi_{\mathrm{LTP}} - \psi_0) \ \% \ 180 \tag{12}$$

The addition of 180 degrees is to ensure that the PA is measured from the local North increasing positively towards the East and the modulo takes care of the 180-degree symmetry.